PicDetect: A Deep Learning Approach for Context-Aware Object Recognition and Risk Analysis

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ABSTRACT

PicDetect is a smart computer vision system that is equipped to recognize objects in real-time and analyze the contextual risk based on deep learning algorithms. The system uses the YOLOv8 object detector and human pose estimator to detect people, weapons, and environmental objects in a picture.

Compared to the traditional object detection systems, PicDetect combines multi-level analysis to include object detection, human pose estimation and scene context understanding. It performs interactions between identified objects including the distance between a human and a weapon and analyzes human pose to identify aggressive behavior by using keypoint-based pose detection.

A scene classification module is a rule-based system used to detect the environment, including an outdoor, kitchen, and office setting, and it is dynamically adjusted to the risk level. Moreover, a hybrid scoring system calculates a normalized score of risk (between 0 and 100) and divides situations into SAFE, LOW, MEDIUM, HIGH and CRITical categories.

In order to enhance resilience, the system uses multi-pass detection schemes and visual heuristic schemes to identify sharp objects when deep learning models miscarry. As it is experimentally proven, PicDetect is an efficient way to combine the accuracy of detection and contextual reasoning and can be used in surveillance, people safety, and intelligent monitoring systems.

Keywords

Computer Vision, YOLOv8, Object Detection, Pose Estimation, Risk Analysis, Deep Learning, Scene Understanding.

I. INTRODUCTION

Conventional object detection solutions are based on recognizing objects in an image without putting them into context or interpreting their interactions with each other. Although these systems are highly accurate, they cannot be used to make sense of the real world e.g. whether an object that has been detected is a threat or not.

In the real world, risk does not only depend on the object itself but on how it is applied and its contact with human beings. To illustrate, a knife lying on a table is not dangerous but when that same knife is placed near an individual in an aggressive pose, then it may point to a risky situation.

In order to overcome this drawback, this paper suggests PicDetect, a contextual intelligent system that combines object detection, human pose estimation, and scene understanding to assess possible dangers in a photograph. The system employs the YOLOv8 features to detect objects and estimate human poses and rule-based reasoning engine to analyze the interactions and contextual factors.

The key contributions of this work are:

•Combination of object detection and pose estimation to analyze the context.
•Creation of a risk hybrid scoring system.
•Scene-sensitive risk level adjustment.
Use of fallback visual heuristics to enhance robustness.

II. LITERATURE SURVEY

In the recent developments in computer vision, detection systems have become much more accurate and efficient in detecting objects. The YOLO (You Only Look Once) family of models is one of the most popular models; they are used to detect objects in real-time by applying the model over an entire image in a single forward pass.

A number of studies have been done to improve object detection performance with the help of deep learning methods. Other models like Faster R-CNN and SSD (Single Shot Detector) have also been found to be highly accurate in their detection but tend to be computationally complex than the YOLO-based models.

Besides the detection of objects, human pose estimation has become an area of interest to human activity and behavior. Combining pose estimation with object detection enables systems to understand human-object interactions in a better way.

The context-aware systems that examine interactions between objects and environmental conditions have also been studied recently. Nonetheless, most of the currently available methods do not generate performance in real-time or do not make efficient use of contextual reasoning.

To overcome these shortcomings, the proposed PicDetect system will integrate object detection, pose estimation and rule-based reasoning to evaluate risks in real-time.

III. PROPOSED SYSTEM

The system proposed, PicDetect, is an intelligent image analysis system that is context-aware and can detect objects in real-time and evaluate their risk. It combines object detection, human pose estimation and scene classification to comprehend interactions among objects.

The workflow of the system involves:

•Image preprocessing

The following are examples of objects that can be detected with YOLOv8.

•Multi-pass detection to enhance accuracy.

•K keypoint-based human pose estimation.

•Rule-based scene classification

•Contextual interaction analysis

•Risk scoring and classification.
•Output visualization

IV. METHODOLOGY

The system is a pipeline of multi-stage pipeline as a mixture of deep learning and rule-based logic.

Object Detection

YOLOv8 identifies objects and gives bounding boxes and labels and scores.

Multi-Pass Detection

Repeated detection steps enhance robustness and decrease false detections.

Pose Estimation

To analyze the posture of a human being and identify an aggressive behavior, keypoints are extracted.

Scene Classification

Rule-based logic recognizes such environments as kitchen or office.

Risk Analysis

The system evaluates:

- Object proximity
- Human posture
- Scene context

Risk Scoring

R = w_1 P + w_2 D + w_3 S

Where:

R = Risk score

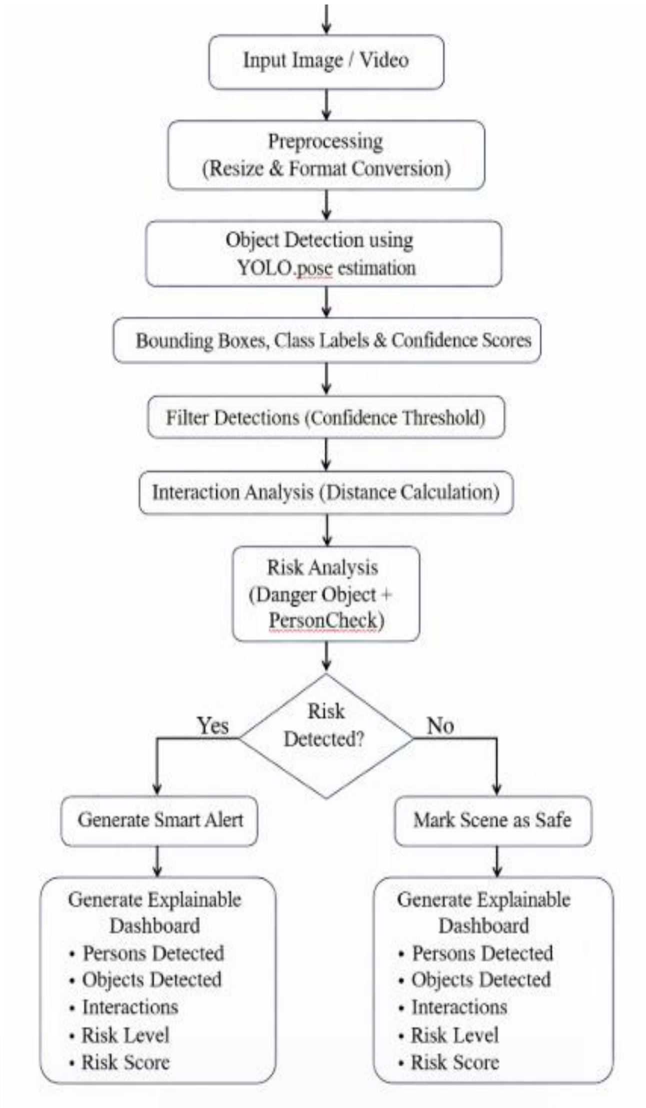
P = Pose factor

D = Distance factor

S = Scene factor

Output

Shows boxes, pose points and ultimate risk level.



V. RESULTS and DISCUSSION.

The system was subjected to different images to test on the detection and risk analysis.

•Proper identification of items such as individuals and guns.

Pose estimation Pose estimation was able to determine aggressive postures.

Context-aware analysis was used to enhance decision-making.

Multi-pass detection enhanced reliability.

Risk scoring was clear in classification.

Constraints are reliance on pre-trained models and rule-based classification of scenes.

VI. APPLICATIONS

(a) Surveillance systems and security systems.

- Smart city monitoring
- Public safety analysis
- Workplace safety
- Intelligent video analytics
- Smart home security.

VII. FUTURE SCOPE

•Trained models to be more accurate.
•Scene classification with deep learning.
•Real-time video processing.

Temporal sequence model analysis.

•Risk scoring based on machine learning.
•Internet of Things and cloud computing.
•Edge deployment optimization

VIII. CONCLUSION

PicDetect is a context-sensitive object detecting model that integrates deep learning with smart reasoning. The proposed system is different as compared to the traditional systems that only concentrate on the identification of objects, but it gives more focus on the relationships among the objects, human behavior, and the environment in order to determine potential risks in a better manner.

The YOLOv8 object detection and human pose estimation help the system to examine both space-related and behavioral data in a picture. The system may distinguish between ordinary and potentially harmful scenarios by adding logic based on rules and situation analysis, which is why it increases the accuracy of decision-making.

Further, a structured risk scoring mechanism is used to report a clear and interpretable representation of the levels of threats and hence the system is more realistic in use in real world applications. Multi-stage processing pipeline increases the detection strength and guarantees the ability to perform fairly in various situations.

The results of the experiment suggest that the system can be used to carry out the correct object detection and significant contextual interpretation. Despite the limitations of the system, including the reliance on predefined rules and pre-trained models, it provides a scalable base of creating more advanced intelligent surveillance systems.

In general, PicDetect fills the gap between object detection and the analysis of the real-life situation and proves to be effective in such applications as security monitoring, the public safety, and the analysis of the smart environment.

X. REFERENCES

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